TECHNICAL CLUB

Audition Tasks

Recruitment 2026 | First, Second & Third Year

01	**Electronics**	Idea + circuit (1st yr), simulation (2nd yr)
02	**Web Development**	Tasks for 1st, 2nd and 3rd year
03	**Graphics**	Pick one design option
04	**Video Editing**	30-35 sec Resume Workshop reel
05	**3D Modelling & VFX**	Original 3D design, pick one approach
06	**Publicity**	Event plan and marketing report

Read this first. Choose your domain, follow its single task format, and finish your work **before** the audition. There is no online submission: you must **bring your task with you on your interview date**, along with the laptop or device you worked on where mentioned.

DOMAIN 01

Electronics

Virtual Electronics Project Challenge | 1st Year: idea + circuit | 2nd Year: full simulation in Tinkercad

OBJECTIVE

Using the component list provided, come up with an electronic project. You need not use every component. Use your creativity to build something useful, innovative or interesting. Think beyond the obvious: turn a simple circuit into a meaningful solution.

RULES (BOTH YEARS)

- Any combination of listed components may be used; using all of them is not compulsory.
- The project must have a clear purpose or functionality.
- Everything is done virtually. No physical components, breadboards or jumper wires.
- Resistors and capacitors are not in the list.
- Innovative applications are encouraged over commonly available projects.
- A component outside the list needs the organizers' permission.

1ST YEAR: IDEA AND CIRCUIT ONLY

Bring your **project idea** and the **circuit / schematic** on the interview date. The **report part must be handwritten**. Code and simulation are not required.

2ND YEAR: FULL SIMULATION

Same task, but you must also simulate it in **Tinkercad**. On the interview date, **bring your laptop** with the **circuit**, the **program / code** (if applicable) and the **working simulation**, plus a **handwritten report** with a brief explanation of the working principle. Be ready to open and demonstrate it.

BRING ON INTERVIEW DATE

- **1st year:** handwritten report (idea + circuit/schematic).
- **2nd year:** laptop with the Tinkercad project, and the handwritten report.

COMPONENT LIST

Component	Qty
Arduino UNO	1
LED	10
RGB LED	1
Push Button	4
Buzzer	1
LDR	2
IR Sensor	2
Ultrasonic Sensor (HC-SR04)	1
PIR Sensor	1
DHT11 Temperature & Humidity Sensor	1

Component	Qty
Potentiometer	2
7-Segment Display	1
16x2 LCD Display	1
Servo Motor	1
DC Motor	1
L293D Motor Driver	1
Relay Module	1
BC547 Transistor	3
1N4007 Diode	4

DOMAIN 02

Web Development

Technical Club Recruitment Task: Website Development | Choose tasks by your year

FIRST-YEAR STUDENTS

Objective: Build a creative and visually appealing website using HTML and CSS, with optional JavaScript. This evaluates your creativity, web development fundamentals and problem-solving skills. Choose **ONE** task.

Task	Requirements
1. Personal Portfolio Website	Create a personal website to introduce yourself and showcase your interests. Sections: Home with your name and introduction, About Me, Skills and interests, Projects or achievements (if any), Contact or social links.
2. Mini Games Website	A simple website with browser-based games: attractive homepage, at least one playable game, clear instructions and result display, a restart button. Suggested: Rock Paper Scissors, Tic-Tac-Toe, Number Guessing Game.
3. Gamified Portfolio Website	Turn a personal portfolio into a game-inspired experience: personal introduction and profile, skills as levels or achievements, a progress bar or achievement badges, at least one interactive feature.
4. Animated Website	Show creativity with CSS animations: attractive hero section, at least two CSS animations, hover effects on buttons or cards, creative typography, colors and layout.

TECHNICAL REQUIREMENTS

- Use HTML5 and CSS3.
- JavaScript is optional, except where required for game functionality or interactive features.
- No frameworks or libraries such as React, Bootstrap or Tailwind CSS.
- Focus on clean code, good design and working features.

LIVE DEMONSTRATION GUIDELINES

- Bring your laptop with your project ready to run.
- Demonstrate your project live and explain its key features.
- Be prepared to explain your code, technology choices and implementation.
- If deployed, you may also showcase the live website.
- Keep your GitHub repository and README ready for reference.

> **SUBMISSION: BRING EVERYTHING ON THE DAY OF AUDITION.** There is **no online submission**: do not send or upload your project anywhere. Bring your laptop with the project ready to run, and keep your GitHub repository, README and deployed link (if any) ready to show.

IMPORTANT NOTES

- Choose only ONE task.
- You are not expected to know everything beforehand. Focus on learning and implementing the basics.
- Use of AI is allowed, but you must be able to debug and change small functionalities (if asked) during the interview.
- A simple, well-executed project is better than an incomplete, complicated one.
- Deployment is encouraged but not compulsory.

All the best! Be creative, experiment, and have fun building your first website.

SECOND-YEAR STUDENTS

Choose any **ONE** of the three tasks.

Task 1: Operation Hidden Code (Frontend Development)

Tech stack: HTML, CSS, JavaScript

Build a futuristic mystery website where users explore an interactive interface, solve puzzles and uncover a secret code.

Mandatory:

- Creative UI with animations and visual effects.
- Interactive terminal with at least five custom commands.
- At least three connected puzzles.
- One hidden Easter egg.
- Final secret-code reveal.
- Feedback for correct and incorrect answers.

Bonus: Timer, scoring system, sound effects, saved progress, or multiple endings.
Bring on audition day: GitHub repository, deployed website and a short demo (all shown live on your laptop).

Task 2: Full-Stack Development (Build Your Own Product)

Tech stack: Open choice (MERN, Next.js, Django or any suitable stack)

Build a complete, functional web application that solves a real-world problem or provides an original user experience.

Mandatory:

- Interactive frontend.
- Backend APIs with meaningful business logic.
- Database integration.
- User authentication and appropriate access control.
- Input validation and error handling.
- At least one feature beyond basic CRUD operations.
- Deployed working application.

Suggested ideas:

- Collaborative workspace with real-time updates.
- Gamified coding platform with leaderboards.
- Campus community platform.
- Hackathon teammate-matching platform.
- An original application idea.

Bonus: AI integration, real-time communication, automated tests, or an admin dashboard.
Bring on audition day: GitHub repository, deployed application, README, and a demo explaining the architecture and technical decisions (all shown live on your laptop).

Task 3: Innovation Project (Think Beyond the Ordinary)

Tech stack: Open choice

Identify a real-world problem and develop an innovative solution using technology. It may involve web development, mobile applications, AI/ML, IoT, automation, cybersecurity or any other relevant field.

Requirements:

- Identify a genuine problem and its target users.
- Propose an original or meaningfully improved solution.

- Develop a working prototype or minimum viable product (MVP).
- Demonstrate the core functionality.
- Explain the technology used and why it was chosen.
- Describe the potential impact, limitations and future scope.

Bonus: A unique technical approach, user testing, measurable results, or a clear plan for real-world implementation.

LIVE DEMONSTRATION GUIDELINES

- Bring your laptop with your project ready to run.
- Demonstrate your project live and explain its key features.
- Be prepared to explain your code, technology choices and implementation.
- If deployed, you may also showcase the live website.
- Keep your GitHub repository and README ready for reference.

> **SUBMISSION: BRING EVERYTHING ON THE DAY OF AUDITION.** There is **no online submission**: do not send or upload your project anywhere. Bring your laptop with the project ready to run, and keep your GitHub repository, README and deployed link ready to show.

GENERAL GUIDELINES

- Choose any ONE of the three tasks.
- Task 1 must use HTML, CSS and JavaScript without frontend frameworks.
- Task 2 and Task 3 allow suitable technologies of your choice.
- Use of AI is allowed, but you must be able to debug and change small functionalities (if asked) during the interview.
- A working, well-explained project is valued more than unnecessary complexity.

Don't just build a project. Build something worth remembering.

THIRD-YEAR STUDENTS: ADVANCED DEVELOPMENT & INNOVATION

Choose **ONE** of the following challenges. Tech stack for all tasks: open choice.

Task 1: Build a Production-Ready Web Application

Design and develop a full-stack application that solves a real-world problem and demonstrates production-level engineering practices.

Mandatory requirements:

- Responsive and user-friendly interface.
- Frontend, backend and database integration.
- Authentication and role-based access control where appropriate.
- Proper API design, validation and error handling.
- Secure handling of user data and sensitive configuration.
- At least one advanced feature, such as real-time updates, background processing, analytics or intelligent recommendations.
- Deployment with a working demonstration.

Bonus: Automated tests, CI/CD, caching, logging, monitoring or containerization.

Task 2: AI-Powered Innovation Challenge

Build an AI-powered application that solves a meaningful problem using AI, machine learning, intelligent automation or data-driven decision-making.

Mandatory requirements:

- Clearly defined problem statement and target users.
- A working prototype with a meaningful AI/ML component.
- Integration with an application interface or workflow.
- Evaluation of output quality using suitable test cases or metrics.
- Handling of invalid inputs, failure cases and limitations.
- Explanation of the model, approach and technology choices.

Bonus: Retrieval-Augmented Generation (RAG), agentic workflows, local models, model evaluation, or cost and latency optimization.

Task 3: Open Innovation (Build Something Impactful)

Identify a real-world problem and create an original technological solution. You may work in domains such as healthcare, education, agriculture, cybersecurity, accessibility, sustainability, smart cities or developer tools.

Mandatory requirements:

- Identify the problem and intended users.
- Propose a solution with a clear innovation factor.
- Build a working prototype or minimum viable product (MVP).
- Demonstrate the core functionality using a realistic scenario.
- Explain the architecture, technology choices and implementation.
- Discuss feasibility, limitations and future improvements.

Bonus: User validation, measurable impact, novel algorithms, hardware integration, or a credible path to real-world adoption.

LIVE DEMONSTRATION GUIDELINES

- Bring your laptop with the project ready to run.

- Demonstrate the working application or prototype live.
- Explain the architecture, key implementation decisions and technical challenges.
- Be prepared to answer questions about your code and show how your solution handles realistic scenarios.
- Keep your GitHub repository, README and deployment link ready for reference, if applicable.

> **SUBMISSION: BRING EVERYTHING ON THE DAY OF AUDITION.** There is **no online submission**: do not send or upload your project anywhere. Bring your laptop with the project ready to run, and keep your GitHub repository, README, and deployment link (if applicable) ready to show.

IMPORTANT NOTES

- Choose only ONE task.
- You may use any suitable programming language, framework or technology.
- Use of AI is allowed, but you must be able to debug and change small functionalities (if asked) during the interview.
- A working, well-tested solution is more valuable than a feature-heavy but unreliable prototype.

The challenge: Don't just demonstrate what you know. Demonstrate what you can engineer.

DOMAIN 03

Graphics

Selection task | Choose ANY ONE of the four options

Option	Brief
1. Logo Design	Original logo for a fictional 12-hour overnight hackathon, **"Hack the Night"**. Submit the final logo with 2-3 lines explaining concept, color palette and font choice.
2. Poster Design	Poster for a fictional college tech fest, **"TechNova 2026"**. Include title, catchy tagline, date, time and venue. Feature the events: Hackathon, Robo Race, Web Design. Size: A4 or 1080 x 1350 px.
3. Social Media Post	Instagram post announcing a fictional AI/Robotics workshop by a college technical club. Include event title, tagline, date, time, venue. Clear AI, robotics or futuristic theme. Size: 1080 x 1350 px.
4. App UI Design	Mobile UI for a fictional college productivity app, **"NexTask"**. Home screen with tasks and upcoming deadlines, plus a task details screen with priority and completion status. Consistent modern, tech-inspired theme. Size: 390 x 844 px per screen.

GUIDELINES

- Your design must be **100% original**. Copying or tracing designs from the internet is not allowed.
- Use any tool: Photoshop, Illustrator, Figma, Canva or Photopea.
- Keep the **editable source file** along with the final PNG/PDF exports, as applicable.
- **Bring your work and the device you made it on** on your interview date.

DOMAIN 04

Video Editing

Technical Club Recruitment Task: Video Editing

THE TASK

Create a creative and engaging **30 to 35 second promotional reel** for our **Resume Workshop** using the footage provided by the Technical Club.

Task sources (Google Drive): https://drive.google.com/drive/folders/1xHSxZrXnPcdCxIdSqCYCEzXLHL81Qi3Y

REQUIREMENTS

At least 3 of the following editing elements must be incorporated. **Sound Effects (SFX) are mandatory.**

- **1. Story Build-up & Narration:** a proper storyline with an engaging beginning, smooth progression and impactful ending. Use narration or voice-over wherever suitable.
- **2. Quick Cuts & Transitions:** well-timed cuts and creative transitions to maintain rhythm and engagement.
- **3. Cinematic Editing:** cinematic visuals, suitable camera angles and colour grading to enhance the overall look.
- **4. Text & Typography:** creative typography, appropriate font selection, animated text and well-timed text transitions.
- **5. Sound Effects (SFX), mandatory:** suitable sound effects for transitions, text animations and visual elements, properly synchronised with the visuals.
- **6. Speed Ramping:** variations in speed to create dynamic and engaging sequences.
- **7. VFX (Visual Effects):** suitable visual effects to enhance the edit wherever required.
- **8. AI-Generated Clips:** AI-generated visuals may be used creatively, provided the original workshop footage remains the primary source.
- Candidates may use suitable music, sound effects, fonts and other editing resources as required.

INSTRUCTIONS

- The provided workshop footage must be the primary source for the edit.
- The reel should promote the Resume Workshop and showcase your creativity and editing skills.
- Experiment with different editing styles while keeping the presentation professional and engaging.

TASK SUBMISSION

> **No submission required.** Bring the device on which you worked on the task to the interview and present your edit.

Note: the objective is to transform the provided footage into an engaging promotional reel that reflects your storytelling, typography, sound design and overall video editing skills.

All the best!

DOMAIN 05

3D Modelling & Visual Effects

Technical Club Recruitment Task: 3D Modelling & Visual Effects

1. OBJECTIVE

Create a unique and creative 3D design based on any concept of your choice: objects, environments, characters, futuristic concepts or imaginative scenes. The focus is on creativity, visual presentation, materials, lighting and bringing your imagination to life.

2. CHOOSE YOUR APPROACH (ANY ONE)

Category	Details
A. 3D Modelling (Blender / Maya)	Create a 3D model or scene for your concept. Focus on modelling, materials, textures, lighting and rendering.
B. 3D Animation (Spline)	Create a 3D animation of your concept. Experiment with motion, transitions, camera movements and interactive elements.
C. 3D Web Development (Three.js)	Create a webpage featuring your 3D design. Explore interactive objects, camera controls, animations and other creative features.
D. AI-Assisted 3D Workflow	Explore AI tools, agents, prompt engineering and prompt chaining to bring your concept to life. You may also try AI-assisted workflows in Blender.

3. DESIGN REQUIREMENTS

- Create an original design based on any concept of your choice.
- Focus on creativity, detailing and visual appeal.
- Use suitable materials, textures, lighting and colours.
- Incorporate animation or interactivity wherever relevant.
- Ensure the final output reflects your own ideas and effort.

4. TOOLS & RESOURCES

- You may use Blender, Maya, Spline, Three.js or other suitable tools.
- Free textures, HDRIs, documentation and online tutorials are allowed.
- AI tools may be used for concept development, prompting and assisting your workflow.
- You must understand your project and be able to explain how you created it.

5. SUBMISSION

- Project submission is **not mandatory**, but you are encouraged to complete your work and prepare a **video walkthrough** of your final output and creative process.
- If you have a project or video, **bring it with you on your interview date** to demonstrate your work.

6. INTERVIEW REQUIREMENTS

- **Bringing the device you used to create your project is compulsory**: your laptop or the relevant device you worked on.
- Be prepared to open and demonstrate your project and explain the tools, techniques and workflow used.
- If the project is incomplete or the video is not made, you may still attend and discuss your approach or show the work completed so far.

Explore. Experiment. Create. Bring your imagination to life in 3D!

DOMAIN 06

Publicity

Bring a written report

OBJECTIVE

Show how you think about promoting and running a technical event.

THE TASK

Prepare **one report** that covers both parts below.

REQUIREMENTS

- **Part A:** A publicity report on the publicity idea of any **past technical club event**: what was done and how it worked.
- **Part B:** A plan for an event **you** would conduct: **what** the event is, **why** you would hold it, **when** and **how** to conduct it, and **how to market it**, from start to finish.

WHAT TO SUBMIT / BRING

- **Bring your report with you on your interview date** (printed or digital).

BRING YOUR TASK TO THE AUDITION

Whichever domain you choose, **you have to bring your task with you on your interview date**. Electronics: handwritten report (2nd years also bring a laptop with the Tinkercad simulation). Web Dev, Graphics, Video and 3D: bring your laptop or the device you worked on. Be ready to explain what you made and why. Good luck!